PERSON / OBJECT TRACKING USING YOLOv8 AND OPENCV

INTRODUCTION:

Object tracking is an important concept in computer vision that focuses on identifying and following a specific object across multiple frames in a video. It is widely used in real-world applications such as surveillance systems, autonomous vehicles, sports analytics, and activity monitoring.

In this project, a real-time object tracking system is developed using YOLOv8 (You Only Look Once) and OpenCV. The system takes a video as input, detects objects in each frame, and continuously tracks them throughout the video using unique IDs.

The main goal of this project is to build an efficient and optimized tracking system that works accurately even when objects move quickly or appear small in the frame.

OBJECTIVES

The primary objectives of this project are:

- To detect objects in a video using a deep learning model (YOLOv8)
- To track a specific object across all frames in the video
- To assign unique IDs to objects for continuous tracking
- To improve performance by optimizing speed and accuracy
- To handle real-world challenges like small objects and fast motion

TECHNOLOGIES USED

➤ YOLOv8

YOLOv8 is a state-of-the-art object detection model developed for real-time applications. It is fast, accurate, and capable of detecting multiple objects in a single frame.

➤ OpenCV

OpenCV is an open-source computer vision library used for video processing, frame handling, and visualization.

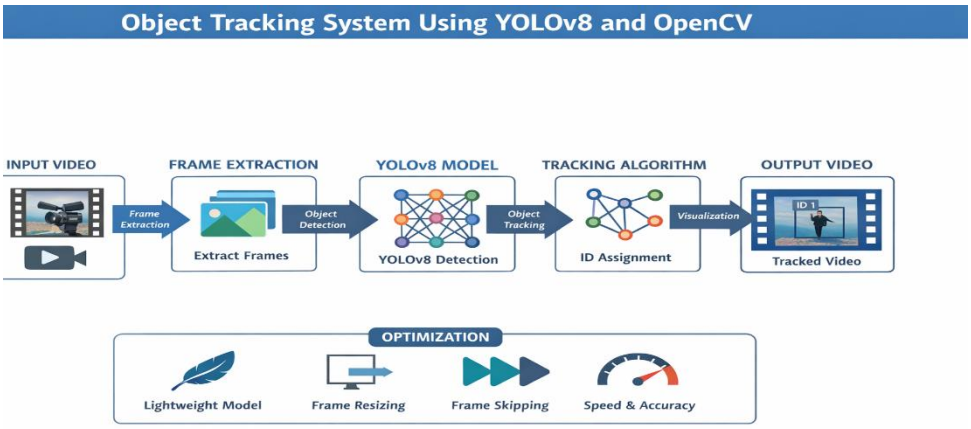
➤ Python

Python is used as the programming language due to its simplicity and strong support for AI/ML libraries.

➤ Ultralytics Library

The Ultralytics package provides an easy interface to use YOLOv8 models for detection and tracking tasks.

SYSTEM ARCHITECTURE



METHODOLOGY

The project is implemented using the following steps:

➤ **Frame Extraction**

The input video is read frame-by-frame using OpenCV.

➤ **Object Detection**

Each frame is processed using the YOLOv8 model, which detects objects and provides bounding box coordinates.

➤ **Object Tracking**

The built-in tracking algorithm (ByteTrack) is used to assign unique IDs to detected objects. This helps in maintaining object identity across frames.

➤ **Visualization**

Bounding boxes and tracking IDs are drawn on the video frames to visually represent tracking

PERFORMANCE OPTIMIZATION

Initially, the system was slow due to heavy model processing. To improve performance, the following optimizations were applied:

- Using a lightweight model (YOLOv8n) for faster inference
- Resizing video frames to reduce processing time
- Skipping unnecessary frames to improve speed
- Adjusting confidence threshold for better detection

These optimizations helped achieve a balance between speed and accuracy.

CHALLENGES FACED

During the implementation, several challenges were encountered:

➤ **Incorrect Classification**

In some cases, objects like skydivers were misclassified as kites due to model limitations.

➤ **Small Object Detection**

Objects appearing far from the camera were not detected properly.

➤ **Slow Processing Speed**

Using larger models resulted in slower video processing.

➤ **Background Noise**

In some frames, only the background was detected instead of the object.

These challenges were handled through model selection and optimization techniques.

RESULTS AND OUTPUT

The final system successfully performs object tracking in video input. The output video shows:

- Bounding boxes around detected objects
- Unique tracking IDs for each object
- Continuous tracking across frames

The system is able to track objects even in dynamic environments with multiple moving elements.

APPLICATIONS

- Surveillance and security systems
- Autonomous vehicles
- Sports performance analysis
- Drone and aerial tracking
- Smart traffic management

CONCLUSION

This project demonstrates an effective approach to real-time object tracking using YOLOv8 and OpenCV. By combining object detection and tracking algorithms, the system is able to monitor objects continuously in a video.

The performance was improved using optimization techniques, making the system suitable for real-time applications. Future improvements can include training custom models and integrating advanced tracking methods for better accuracy